

GOVINDA RAI

PRINCIPAL QUALITY ENGINEERING ARCHITECT • AI-ENABLED QA MANAGER

Quality Engineering leader with 15+ years of experience driving QE transformation through AI-assisted testing, intelligent automation, and modern quality practices across Automotive, Telecommunications, and Data Platform domains. Proven record of leading distributed QA organizations of 20+ engineers while accelerating delivery quality, engineering productivity, and release predictability.

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CAREER SNAPSHOT

Associate Principal Engineer	Nagarro Services Pvt Ltd · Cox Automotive	07/2021 – Present
Test Manager	Vodafone Shared Services (VOIS) · Telecom, SAP & Data	01/2017 – 07/2021
Team Lead / Senior Software Engineer	Accenture Services · Telstra, Vodafone, Cox	08/2010 – 01/2017

CORE COMPETENCIES

▪ QE Strategy & Transformation	AI adoption roadmaps, test governance, TCoE establishment, shift-left and quality-first culture
▪ Leadership & Delivery	Global teams of 20+ engineers, estimation, release planning, resource management, risk mitigation, mentoring
▪ AI-Driven Quality Engineering	Generative AI for test design, automation development, defect triage, RCA, code review and documentation
▪ Automation Architecture	Scalable, reusable frameworks across web, API, mobile, cloud and microservices platforms
▪ Performance & Reliability	Load, stress and scalability engineering for APIs and UI at enterprise scale
▪ Metrics & Business Impact	Automation ROI, traceability, test observability, quality dashboards and executive reporting

PROFESSIONAL PROFILE

Experienced in leveraging Generative AI, GitHub Copilot, and Claude to accelerate test design, automation development, defect analysis, test data preparation, and overall SDLC efficiency — modernizing Quality Engineering through AI-driven testing strategies, intelligent automation, and continuous quality practices.

Hands-on technical leader with deep expertise in Selenium, Playwright, Tricentis Tosca, UFT, API automation, performance testing, microservices validation, and CI/CD integration — and a trusted partner to Product Owners, Architects, and Engineering Leaders in combining QA strategy, automation roadmaps, test governance, and Agile delivery discipline to build high-performing teams aligned to business and technology objectives.

TECHNICAL SKILLS

AI & Generative AI	GitHub Copilot, Claude AI, AI-assisted test design and test data generation, prompt engineering, AI-driven defect triage and RCA, AI-assisted code review and observability
Languages	C#, Java, JavaScript, Python, SQL
Automation & Testing	Selenium WebDriver, Playwright, Tricentis Tosca (incl. Tosca DI), UFT, Appium, Perfecto, Mobile Centre, SpecFlow (BDD), NUnit, TestNG, Keyword-Driven and Hybrid frameworks
APIs & Microservices	REST, SOAP, JSON, Web Services, microservices validation, RestSharp, Postman, Swagger, SOAP UI
Performance Engineering	Apache JMeter, k6, NeoLoad, Google Lighthouse
Cloud & Infrastructure	AWS (ECS, S3, SQS, ELS, CloudWatch), Terraform, Google Cloud Platform
Data & Databases	SQL Server, MongoDB, BigQuery, Airflow DAGs, Data Composer, data quality validation
Observability	New Relic, Splunk, AWS CloudWatch, Grafana, test observability
CI/CD & Tooling	Jenkins, GitHub Actions, TeamCity, Git, JIRA, Rally, Micro Focus / HP ALM, CTDA, ConnectAll, Qlik, Confluence, SharePoint, Visual Studio Code
Methodologies	Agile / Scrum, DevOps, shift-left and risk-based testing, test governance, automation ROI and traceability

PROFESSIONAL EXPERIENCE

Associate Principal Engineer — Nagarro Services Pvt Ltd

07/2021 – Present | Pune, India

Client: Cox Automotive — world's largest automotive services and technology provider

Engagement: Price Guides, Media & Inventory Services

Automotive dealer platforms supporting vehicle valuation and pricing, inventory management, digital retail, and new/used car sales operations — web UI integrated with microservices and APIs hosted on AWS.

Technology Stack: Playwright, Selenium WebDriver (C#/Java), SpecFlow, NUnit, RestSharp, REST APIs, Web Services, Microservices, GitHub Copilot, Claude AI, GitHub Actions, AWS ECS/S3/SQS/CloudWatch, Terraform, Splunk, New Relic, JMeter, k6, NeoLoad, Postman, Swagger, Google Lighthouse, DevOps CI/CD

Quality Engineering Leadership & Transformation

- Lead a 20+ member global Quality Engineering organization across onsite and offshore locations, driving AI-enabled testing, intelligent automation, API validation, performance engineering, and release quality for large-scale enterprise programs.
- Own Quality Engineering strategy, AI adoption initiatives, test governance, release planning, estimation, resource management, risk mitigation, and quality metrics — ensuring predictable, high-quality delivery outcomes.
- Established the Testing Center of Excellence and drove QE transformation and DevOps integration across teams, covering architecture governance, automation ROI, automation traceability, test observability, and business impact metrics.
- Architected and governed scalable automation frameworks using Playwright, Selenium, SpecFlow and C#, establishing reusable standards that improved automation reliability, maintainability, and execution efficiency.
- Directed automation, API, integration, performance, and scalability validation across web, cloud, and microservices platforms using RestSharp, JMeter, k6, NeoLoad and Google Lighthouse — expanding regression coverage and ensuring resilient, production-ready systems.
- Mentor and develop QA engineers in modern Quality Engineering practices, AI-assisted development, prompt engineering, and automation architecture — building high-performing, future-ready teams.
- Partner with Product Owners, Architects, Engineering Leaders, and Business Stakeholders to embed a quality-first, shift-left engineering culture across the software delivery lifecycle.

AI-Driven Quality Engineering

- Spearheaded organization-wide adoption of Generative AI and AI-assisted testing practices, accelerating test design, automation development, defect analysis, and SDLC productivity across delivery teams.
- Apply GitHub Copilot, Claude, and Generative AI tooling to accelerate automation development, framework enhancement, code refactoring, API automation, and test asset creation.
- Use AI for requirement analysis, test scenario generation, defect triage, root cause analysis, debugging, code reviews, and technical documentation — materially improving engineering productivity and delivery velocity.
- Combine New Relic, Splunk, and AI-assisted analysis of production logs, telemetry, and failure patterns to expedite issue diagnosis and defect resolution, and to enable proactive production support.
- Drive AI-assisted infrastructure validation and cloud deployment reviews, including Terraform configuration generation and verification.

Test Manager — Vodafone Shared Services (VOIS)

01/2017 – 07/2021 | Pune, India

Client: Global Shared Services and Technology — global telecom, frontend and SAP applications

Engagement: EVO Automation, Test Automation War Room, VF Portugal CRM

Quality Assurance and test automation leadership for a large-scale cloud-based data platform supporting telecom and SAP ecosystem integrations — establishing data quality validation frameworks, automation strategy, test planning, and release assurance across complex data transformation pipelines on Google Cloud.

Technology Stack: Tricentis Tosca / Tosca DI, Google Cloud Platform, BigQuery, Airflow, Data Composer, SQL, Python, Selenium WebDriver (Java), Maven, REST, JIRA, CTDA, Micro Focus ALM, Qlik, Grafana, Jenkins, SKDF, Test Shell

- Led the QA function for cloud-based data transformation and integration programs, ensuring end-to-end quality across data pipelines, APIs, reporting platforms, and downstream business applications.
- Defined test strategy, test plans, estimation, release validation approach, and quality metrics across multiple parallel delivery streams, governing defect management, regression automation, execution reporting, and release readiness through JIRA and Micro Focus ALM.

- Established automated validation frameworks using Tricentis Tosca for data integrity testing, including DB-to-DB, file-to-file, source-to-target, and transformation validation scenarios.
- Directed validation of Google Cloud data platforms — BigQuery datasets, Airflow DAGs, Data Composer workflows, and complex processing pipelines — alongside API and integration testing that ensured reliable data exchange across interconnected enterprise systems.
- Developed Python-based utilities and test data solutions to support complex validation scenarios and improve testing efficiency.
- Oversaw dashboard and reporting validation using Qlik, ensuring accuracy, completeness, and consistency of business-critical insights.
- Assessed automation maturity and ROI on automated assets, drove best-practice adoption, and delivered tool integration across JIRA, ALM, Tosca and UFT using ConnectAll.
- Conducted feasibility assessments, POCs and solutioning; owned scripting and maintenance, KEDB upkeep, Grafana-based reporting, and configuration of Jenkins pipelines and jobs for test execution.
- Managed and mentored QA team members across functional testing, automation, data validation, and release verification, collaborating closely with Product Owners, Data Engineers, Architects and Development teams to surface risks early and ensure successful delivery.

Team Lead / Senior Software Engineer — Accenture Services Pvt Ltd

08/2010 – 01/2017 | Chennai, India

Client: Telstra (Australia), Vodafone India, Redknee, Cox Communication (US)

Engagement: UI applications and services for the largest telecom provider in the US

Telecom web applications and services focused on end-to-end business workflow validation, automation testing, and quality assurance for customer and communication management systems.

Technology Stack: Selenium WebDriver, Java, Maven, SOAP UI, Micro Focus ALM, Jenkins, Inbill product over Apex, PyCharm, Siebel, Salesforce, XML, TOAD, Kenan

- Led end-to-end Quality Assurance and release validation for telecom web applications and services, ensuring seamless customer management and communication workflows across multiple releases.
- Owned automation strategy, planning, design and script delivery across projects, managing resources and mentoring engineers on automation best practices, framework adoption, and quality engineering process.
- Defined test strategy, test planning, estimation, quality gates, and release readiness activities in partnership with Product Owners, Developers and Business stakeholders.
- Designed and implemented Selenium WebDriver automation frameworks using Keyword-Driven and Hybrid approaches, improving regression coverage and reducing manual testing effort.
- Directed functional, integration and regression testing as Defect Manager for the team, owning defect triage, risk assessment and quality reporting through Micro Focus ALM.
- Performed end-to-end testing in the telecom domain, reporting daily quality metrics to senior management and leading client calls for project and delivery activities.

EDUCATION

Bachelor of Technology, Information Technology — 82%

Heritage Institute of Technology, Kolkata, India | 08/2006 – 05/2010

Higher Secondary (WBHS), Science — 82%

Shree Jain Vidyalaya, Kolkata, India | 04/2004 – 05/2006

CERTIFICATIONS & RECOGNITION

Training & Certifications

- ISTQB Certified — Foundation Level
- Tricentis Tosca Test Architect Level 1
- Certified Scrum Master (CSM)
- Micro Focus ALM / UFT Certified
- UiPath & Automation Anywhere Certified
- AI & GitHub Copilot

Awards & Achievements

- Brilliant Mind Award — Nagarro (2022)
- A-Team Award — Nagarro (2022)
- Winner & Gold Medallist — India Testing League 2020
- Winner — TestAutothon 2018

Languages: English and Hindi — full professional proficiency

Interests: Cricket, travelling, snooker

Volunteer: Organizer of fests and interviews, Kolkata (2010)